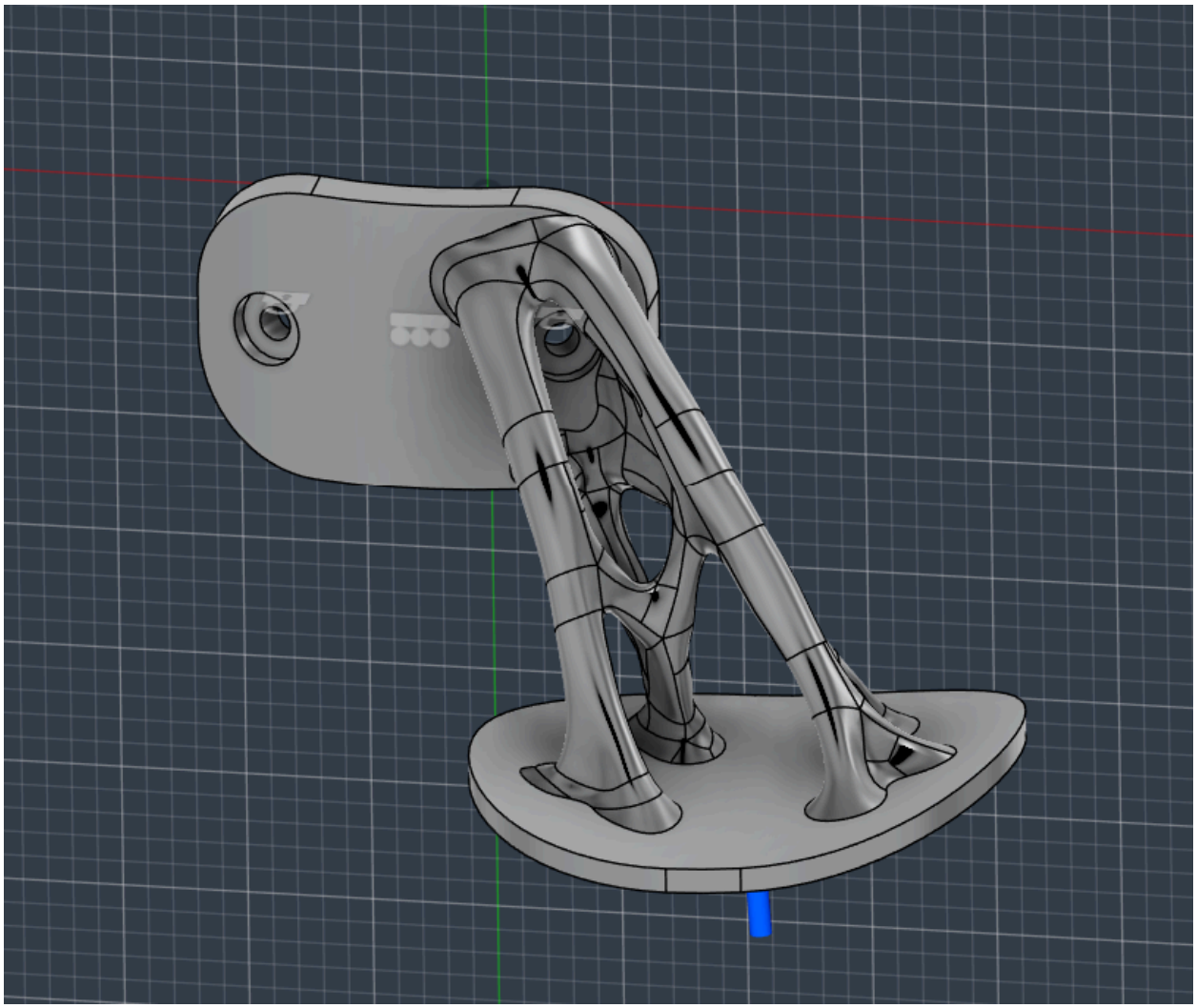
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ME 108 Report - Structural Optimization of an Ergonomic Clarinet Thumbrest



Introduction

The clarinet is a woodwind instrument that presents a unique and demanding ergonomic challenge for the performer. Weighing approximately 800 grams, the entire mass of the instrument is usually supported by a single point of contact located on the right thumb of the musician. Historically, the adjustable thumb rest has been manufactured as a solid simple casting

made from brass or nickel-silver alloys. While these traditional parts are durable, they are often designed without regard for structural optimization.

This report documents the development and simulation of a redesigned thumb rest that utilizes Autodesk Fusion's Generative Design to create a high-performance, aftermarket structural skeleton. The primary goal of this project is to replace the bulky, solid design of traditional rests with a modernized architecture that maximizes rigidity and weight reduction while providing a more ergonomic design. Specifically, we established an optimization objective to minimize mass, subject to a strict 0.1 mm maximum displacement constraint and a minimum Factor of Safety of 2.0. By focusing on an L-bracket style optimized for additive manufacturing, this study seeks to prove that advanced computational design can produce a superior ergonomic and structural alternative.

Background

The necessity for this extended, cantilevered geometry is deeply rooted in the biomechanics of professional woodwind performance. Sustained static loading of the instrument's weight frequently leads to neuromuscular issues, specifically within the carpometacarpal joint of the right thumb. Research published in the journal Medical Problems of Performing Artists by Kathryn Young and Sara Mannello proves that the specific geometry and position of standard thumb rests create high-pressure concentrations, which act as critical factors in chronic hand tension and technical fatigue. By increasing the contact area and shifting the platform outward away from the clarinet body, our redesign directly addresses these clinical findings. This wider, offset base provides a more favorable distribution of force, effectively bypassing the vulnerable joint entirely.

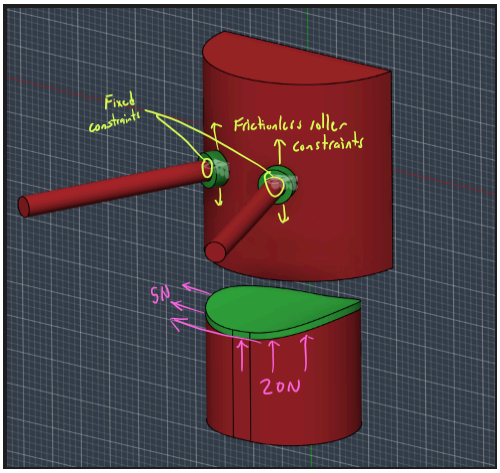
Moreover, the choice of material for this study is important for the professional musician. While various polymer and rubber alternatives exist in the consumer market, they often lack the rigid feel required for high-level technical precision. Professional musicians require hardware that does not flex or vibrate independently of the instrument. To meet these demands, we decided to choose AlSi10Mg, an aerospace-grade additive aluminum alloy, as the design material. This alloy is not typically standard in the woodwind industry; however, it is remarkably lightweight, has a high stiffness, and is often used for metal 3D printing. With an elastic modulus of

approximately 70 GPa and a yield strength of 240 MPa, it provides the structural stability required for a professional-grade accessory. Similarly, the structural advantages of combining lightweight aluminum alloys with computational optimization are well documented in broader mechanical applications. In a design optimization study by A.S. Adkine et al., the authors demonstrated how aluminum alloys—which are one-third the density of steel—provide an excellent stiffness-to-weight ratio when optimized for load-bearing brackets. While their specific application was an engine support bracket, the underlying engineering principle of using optimization software methods to minimize the mass of a cantilevered structure directly translates to our L-bracket design. By applying this same material strategy to our uniquely loaded thumb rest, we ensure the optimized part can maintain its shape under the significant dynamic forces of performance while avoiding the heavy bulk associated with traditional hardware.

Method



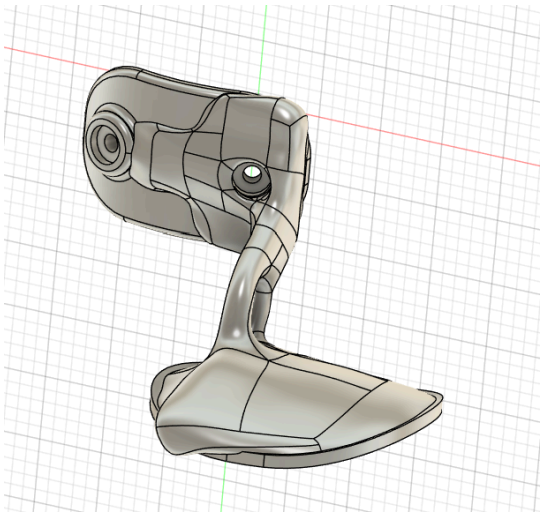




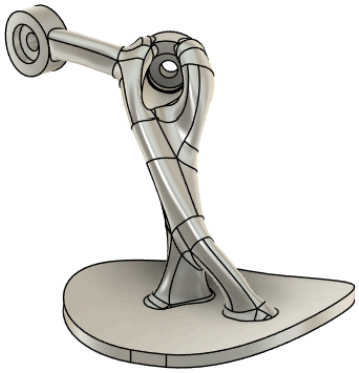
The methodology for this project utilized the Generative Design workspace in Autodesk Fusion to organically build a structure between isolated functional interfaces. Rather than carving material away from a solid bulky block, we began by modeling the essential functional components as completely separate bodies: a bracket for attaching to the instrument with screw standoffs and an extended, ergonomic thumb ledge. Reference to traditional designs and measurements against an actual clarinet were made to determine rough dimensions. These independent bodies were designated as Preserve Geometries (in green), representing the absolute minimum anchor points the software was required to connect.

To ensure the generated geometry would function in the real world, Obstacle Geometries (in red) were defined in three critical areas: in and around the screw holes to maintain screwdriver access, along the lower surface of the ledge to ensure unobstructed thumb clearance, and a large circular bore representing the wooden clarinet body itself. The boundary conditions were defined based on the physical assembly of the instrument. Fixed geometry constraints were applied to the interior cylindrical surfaces of the two mounting standoffs, simulating the rigid anchor points of the screws. A frictionless slider constraint was applied to the back surfaces resting against the clarinet body obstacle. The loading scenario consisted of a primary upward vertical load of 20 N applied to the thumb ledge (simulating a 2.5x dynamic multiplier of the instrument's weight) and a secondary lateral shear load of 5 N to simulate the angled pressure exerted by the thumb in the high register. The optimization objective was set to minimize total mass, constrained by a global displacement limit of 0.1 mm and a minimum Factor of Safety of 2.0.

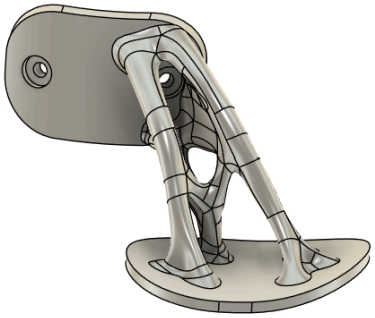
Results



Our initial generative runs produced a slightly heavier, unrefined structural skeleton with a mass of over 4.119 grams. While this iteration was less mass-efficient, its thicker design made it viable for traditional casting. This presented a valid alternative that would potentially allow the use of traditional acoustic materials, such as Nickel Silver 65-12, while still maintaining our design's advantages compared to standard hardware.

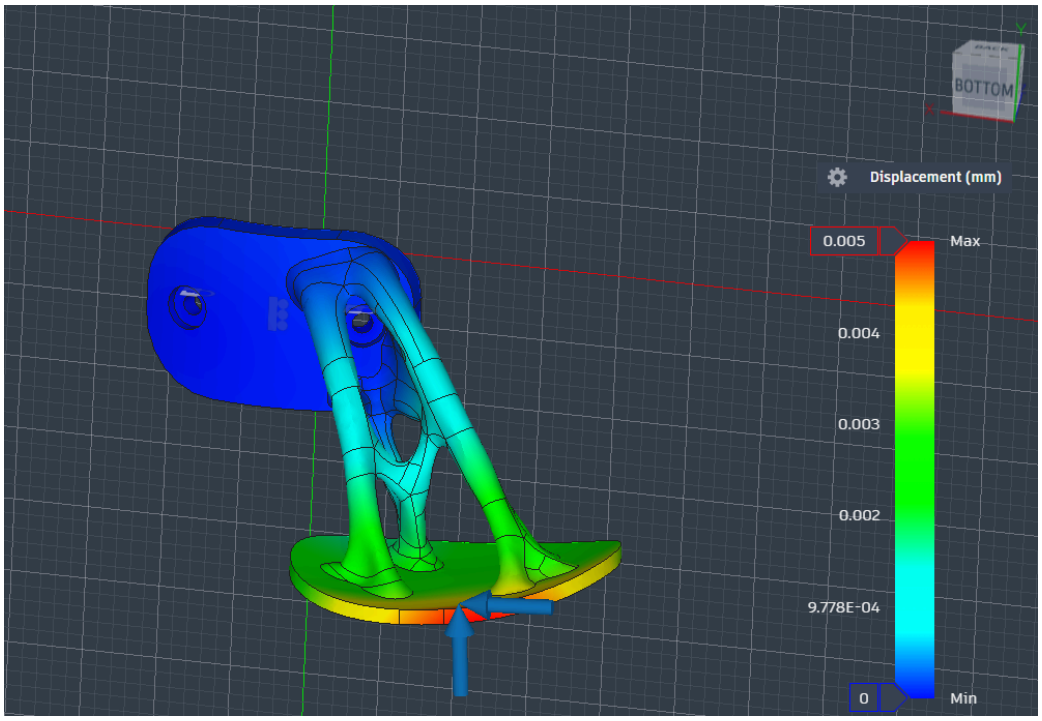


In our second iteration, we reduced the preserved geometry bracket to just the two screw standoffs with minimal material before running the study and got a resulting part with half the mass of the previous iteration at 2.022 grams. While this was highly desirable and still maintained a hefty safety factor, we realized that this design, due to its lower surface area in contact with the instrument body, would likely put significant localized pressure concentrations on the wooden clarinet body itself, risking tearing out the screws.

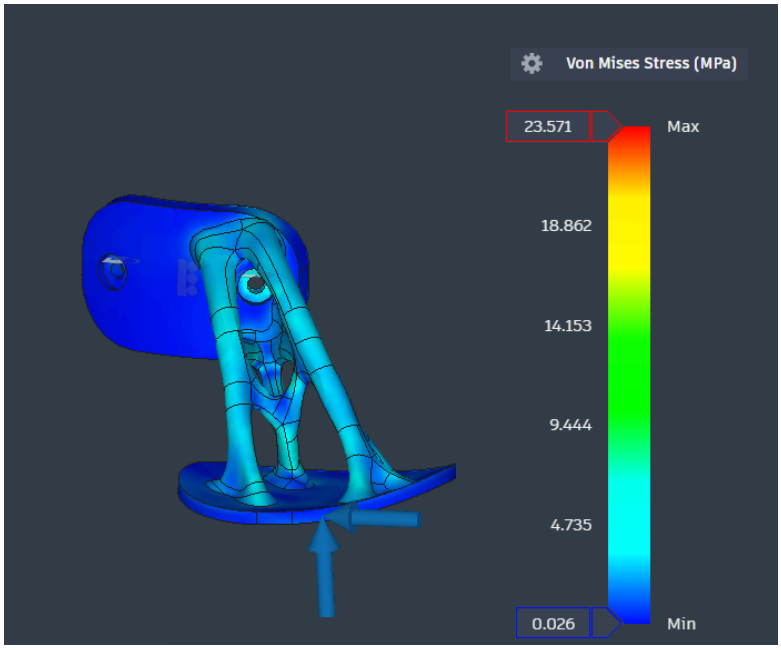


For our final iteration, our critical area of interest was the footprint of the bracket resting against the clarinet body. In our 2.022g iteration, the surface area was reduced to just the two small screw standoffs, which concentrated the entire reaction moment into a tiny footprint. By switching to the full backplate in our final design, we drastically increased the contact surface area, quantitatively reducing the localized pressure on the wooden instrument body to prevent

structural tear-out. We ended up with a much more visually pleasing design with a relatively low weight of 3.086 grams.



A specific analysis of the displacement plot provides the most compelling evidence for the success of this generative design. Under the combined 25 N loading scenario, the maximum displacement recorded on the leading edge of the thumb platform was a mere 0.005 mm. This performance exceeds the initial requirement of 0.1 mm by an order of magnitude, indicating that the tripod-like structure is exceptionally efficient at maintaining structural stiffness. This minute displacement ensures that the musician will perceive the rest as completely stationary, fostering the confidence needed for technical play. Finite element analysis further verified that the maximum Von Mises stress was recorded well below the 240 MPa yield strength of the aluminum alloy, successfully achieving a robust factor of safety. The generative design proved that by distributing material specifically along optimized load paths, a wider and more ergonomic interface can be provided without the weight penalty of traditional solid designs.



Conclusion

This project successfully demonstrates the application of generative design in modernizing specialized woodwind hardware. By transitioning from over-engineered solid castings to a skeletonized AlSi10Mg architecture, we developed an ergonomic thumb rest that significantly reduces material volume while improving technical comfort for the musician. The minute 0.005 mm displacement confirms the design easily exceeds professional-grade stiffness and safety requirements. A significant challenge during the optimization process was balancing the software's drive to minimize mass against real-world mechanical interfaces. While the algorithm generated a highly efficient 2.022g structure, engineering intuition was required to realize its small footprint would create pressure concentrations that could damage the wooden clarinet body. This prompted the enlarged, load-distributing backplate of our final 3.086g design. Because FEA results indicate a massive surplus of stiffness, the part remains well under its structural limits. We could have further reduced mass in the connecting struts as well as the bracket itself without risking instrument damage. Additionally, while AlSi10Mg proved excellent for extreme lightweighting, we recognize that aerospace-grade aluminum is somewhat overkill and deviates from traditional instrument manufacturing. For future work, we suggest adjusting the generative constraints to optimize for investment casting using traditional acoustic materials like Nickel Silver 65-12. Finally, we recommend physically 3D printing a prototype of the

current iteration to practically validate the ergonomic comfort of the shifted ledge against the performer's hand.

References

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